MATHEMATICAL THERMAL MODEL PERFORMANCE REPORT

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PURPOSE

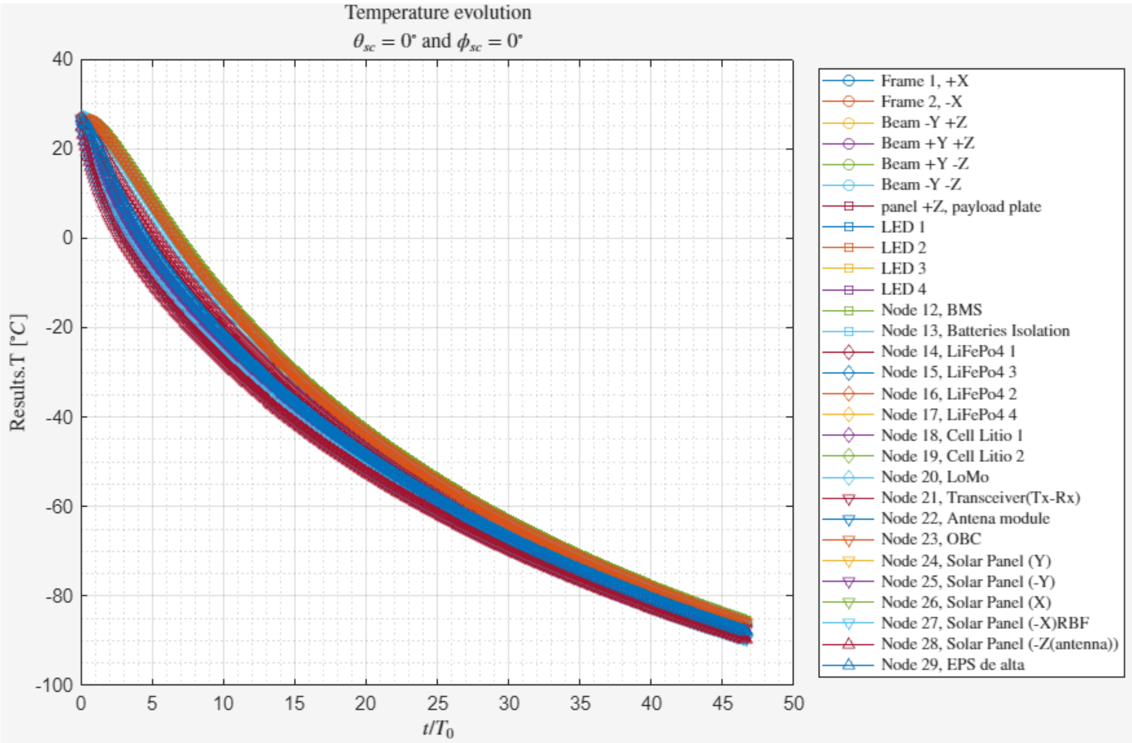
This document aims to compile the tests performed on the mathematical thermal model to review its performance as well as their respective conclusions.

QUALITATIVE TESTS

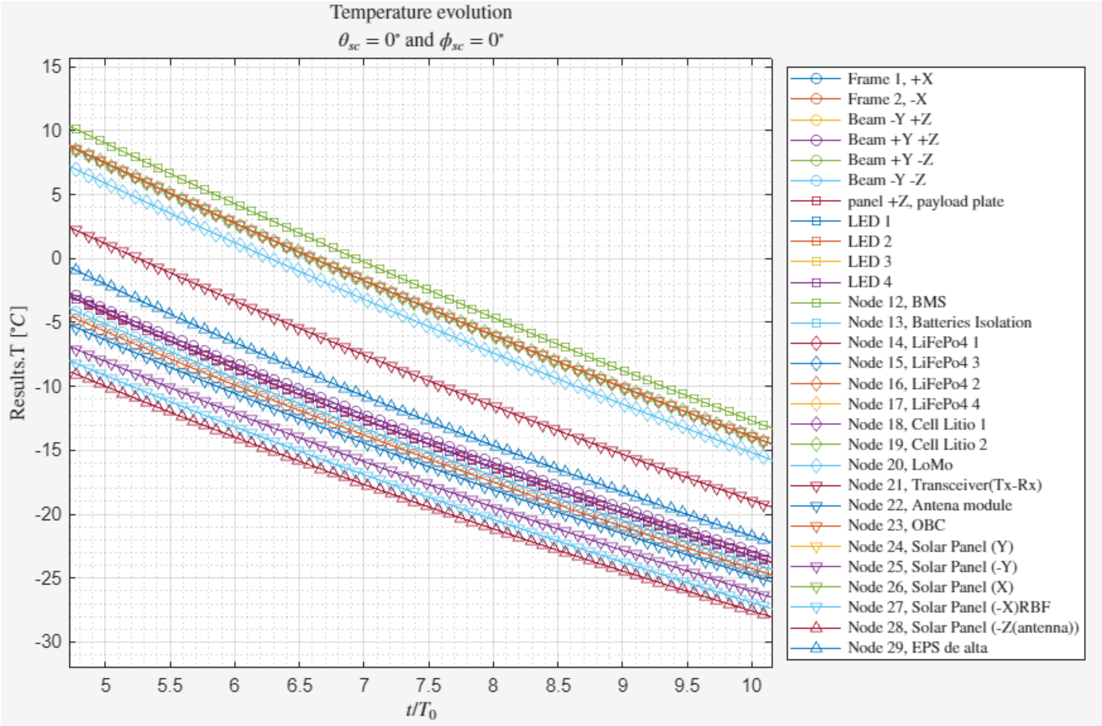
Tests performed to check the general behaviour of the model without taking into consideration the exact numerical value of the results. Only the general tendencies are observed in these tests.

COOLING

The satellite was simulated with all systems off in a completely void and cold environment. An initial temperature of 300K was set.



As expected, the temperature dropped due to radiation emission, at a rate that slowed down as the satellite cooled given the non-linear relation between temperature and heat dissipation.



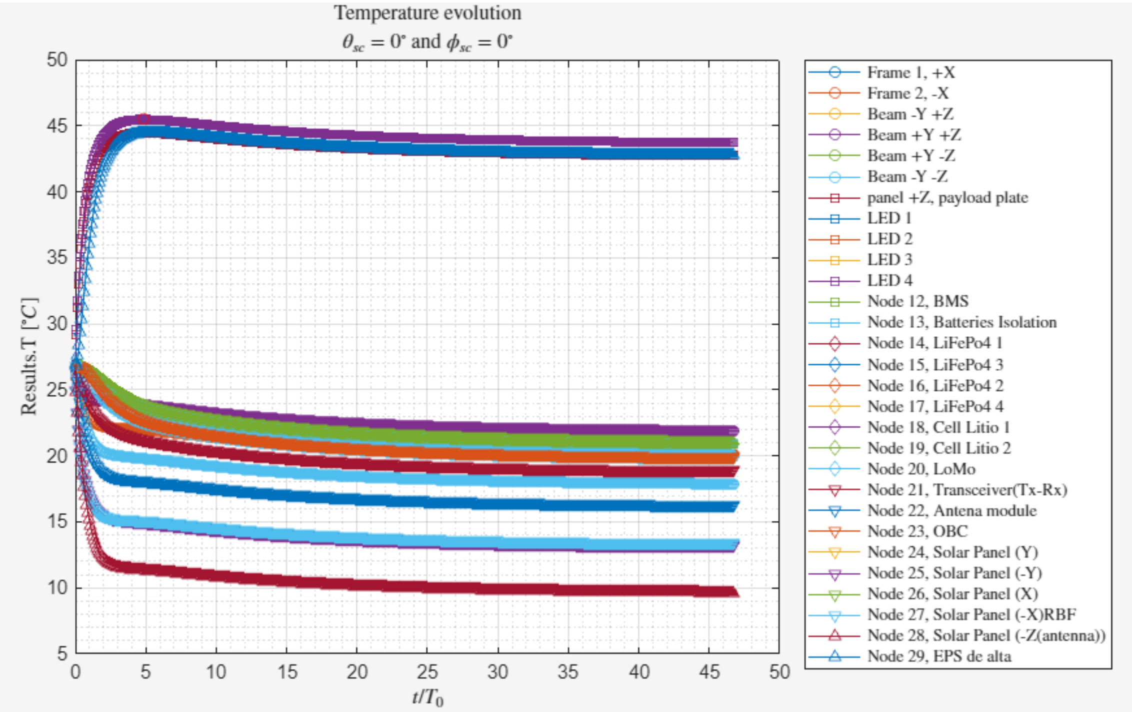
Zooming in, we can also observe that nodes with more surface to radiate, such as the solar panels, are always at a lower temperature than internal components such as the on-board computer (OBC) or the battery management system (BMS). The temperature gradient is to be expected as heat flows to the nodes with higher dissipation capabilities.

This is consistent with theory.

HEATING AND STATIONARY REGIME

The satellite's LEDs were turned on permanently in a cold void environment. The total heat generation was set 24 W.

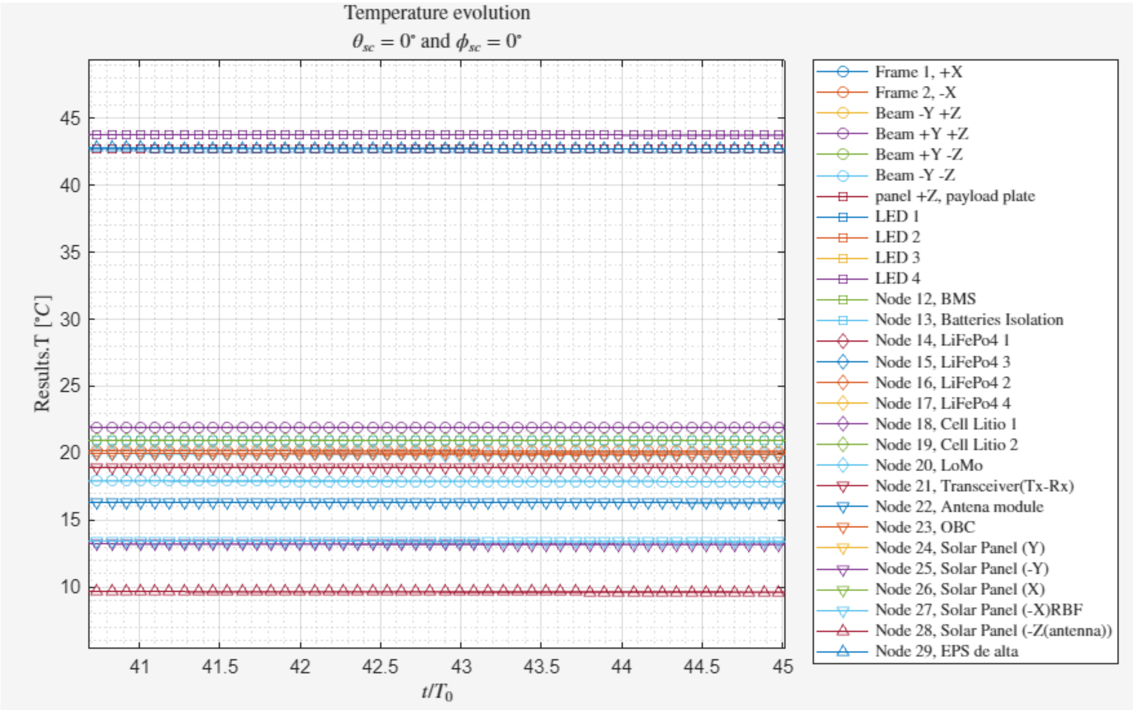
STATIONARY TENDENCY



The temperatures fluctuate before asymptotically approaching some fixed values. This is expected of a transition into stationary regime, approaching the values at which the net heat transfer to every node is equal to zero, and the total radiated heat equals the generated heat for the whole satellite.

This is consistent with theory

INTERNAL CONDUCTION



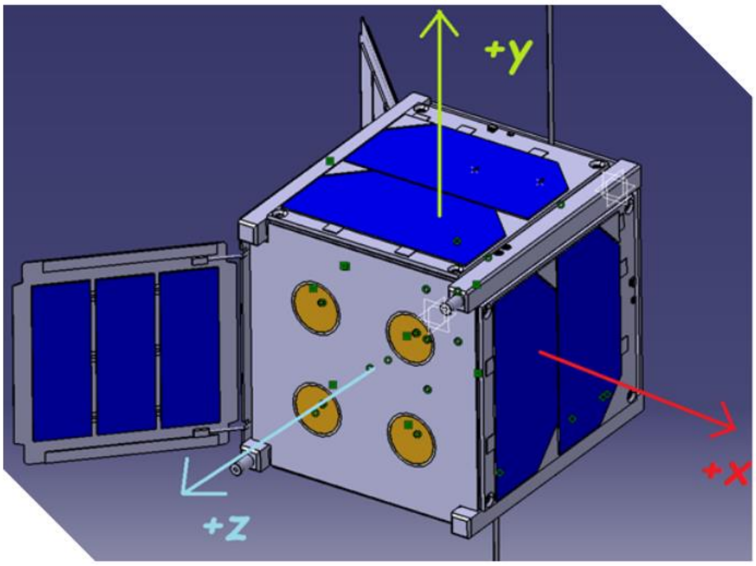
Inspecting the temperatures closer, we can identify that reached stationary regime the temperatures align such that they increase as the components get closer to the heat generating elements (The LEDs) with the LEDs being at the highest temperature. This is consistent with theory as the heat flows from higher to lower temperature.

We can see that the nodes separate into two groups, with the nodes integrated into the payload plate at considerably higher temperatures than the rest of the satellite. This indicates that the conductance between the payload plate and the rest of the satellite is considerably low.

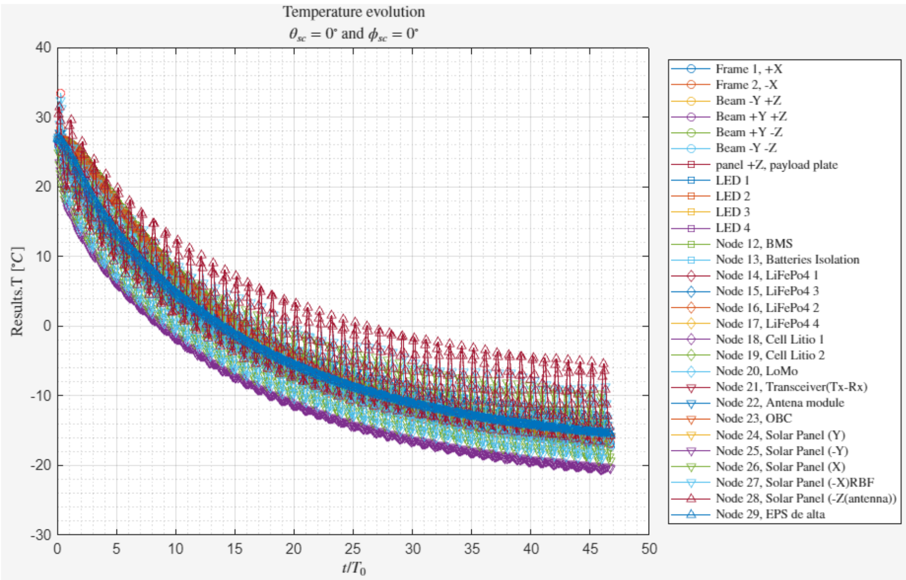
Regardless, this behaviour is consistent with theory.

SOLAR CYCLE AND EARTH RADIATION

With the satellite off, we simulate the sun radiation, albedo and infrared emission from the earth. The orbital parameters chosen define a circular orbit on the ecliptic plane (plane on which the solar system's planets are). The orientation chosen is the default, with the +Z axis of the satellite pointing to earth and the +X to the orbital prograde direction.



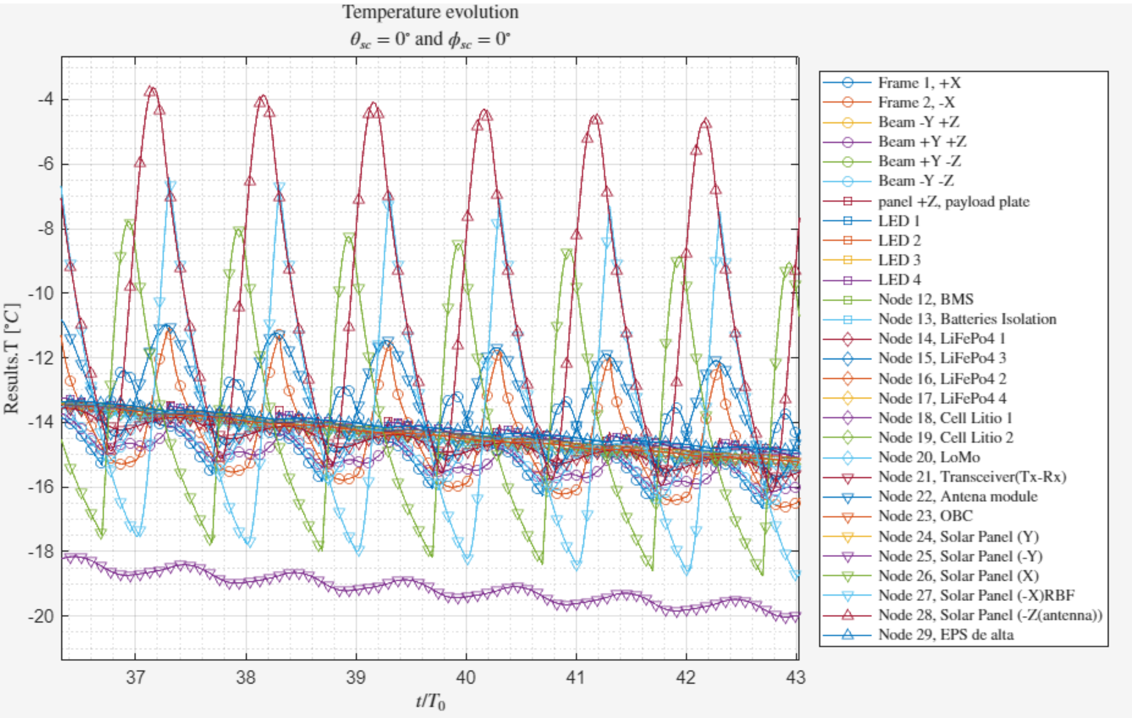
EQUILIBRIUM TENDENCY



We can observe how the satellite's overall temperature decreases asymptotically approaching equilibrium with the environment, when the energy radiated per solar cycle equals the energy absorbed.

This is consistent with theory.

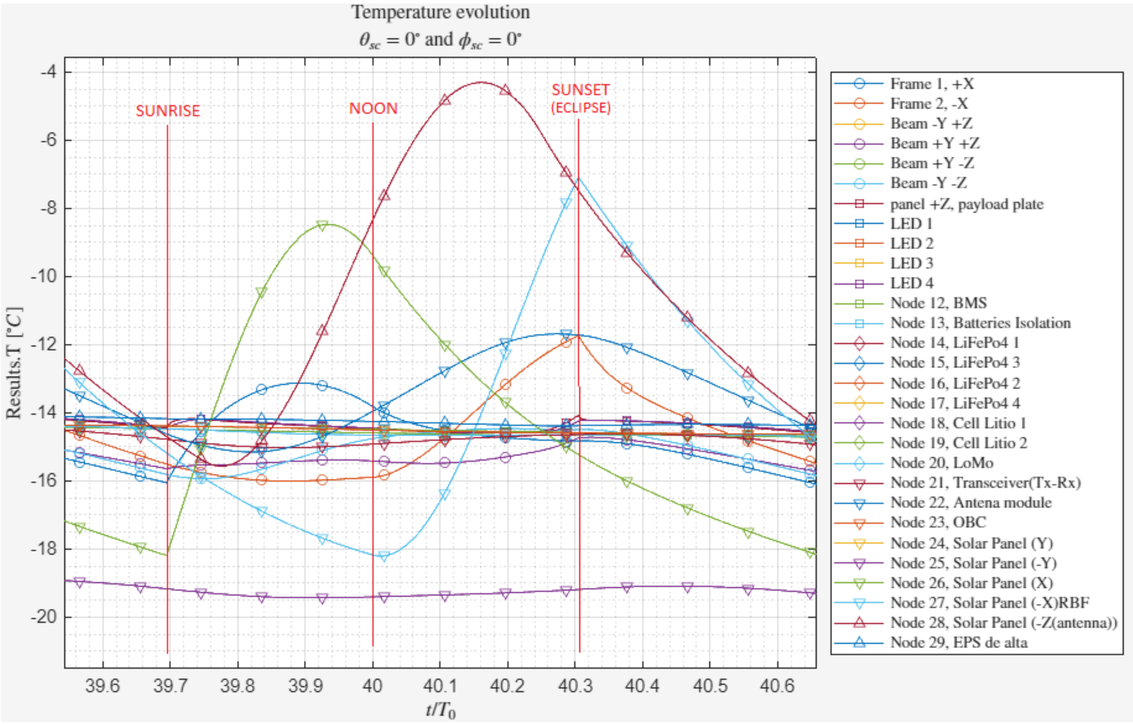
TEMPERATURE FLUCTUATIONS



A closer inspection reveals that those nodes closer to the surface (Solar panels, rails, antennas…) fluctuate with higher amplitude than those closer to the centre (Batteries, computers…). This is expected as the surface nodes are more exposed to the fluctuating outside environment.

This is consistent with theory.

SOLAR CYCLE



We can further test the solar cycle by a detailed observation of the different values. In this orbit, the satellite is expected to experience a view of the sun like the one seen on earth near the equator just with much higher frequency due to the short orbital period. The three important nodes to observe during this analysis are the X,-Z and -X solar panels (The high amplitude lines in green, red and blue respectively). These are the solar panels that will be facing the sun at some point during an orbit with these parameters.

As the satellite leaves the earth shadow, the solar radiation directly illuminates the X solar panel, which we can see as a sharp increase in temperature on the green line.

Progressively, as the satellite rotates as it orbits the earth, the solar radiation starts illuminating the -Z solar panel and reduces the intensity on X. We can observe this as a slow decrease of the temperature slope on X with an increase of the temperature in -Z. This continues until the -Z face is completely facing the sun and the temperature on X decreases due to radiation emission.

As the satellite continues to rotate, the solar radiation starts decreasing over -Z and increasing over -X, which we can again see as slow temperature fluctuations until the -X face receives all the radiation while -Z cools.

Finally, the satellite enters the shadow of the earth again which translates into a sudden sharp decrease in in the -X solar panel.